Hamza Shahbaz



Computer Science Researcher | AI & Deep Learning

Sangla Hill, Punjab, Pakistan • Pakistani

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Google Scholar | ORCID | GitHub | Kaggle

PROFESSIONAL PROFILE

Computer science researcher with an MS in Computer Science, specializing in explainable AI, medical image analysis, computer vision and reproducible machine learning. Research across retinal and MRI imaging, NLP, crop health and scalable cloud-based learning systems.

EDUCATION & TRAINING

MS COMPUTER SCIENCE | Government College University Faisalabad | 2024–2026

CGPA 3.02/4.00 • Thesis: Diabetic Retinopathy Detection on Fundus Images Using Deep Learning.

BS COMPUTER SCIENCE | Government College University Faisalabad | 2019–2023

CGPA 3.27/4.00 • Final-year project: Clinic Management System.

HSSC (INTERMEDIATE) | Computer Science | Government Islamia Postgraduate Degree College Sangla Hill | 2017–2019

SSC (MATRICULATION) | Science | Al Falah Group of Colleges Chak Jhumra | 2015–2017

OFFICIAL HEC

Verify MS degree

Ref: HEC/A&A/DAS/2026/5487117

RESEARCH EXPERIENCE & CONTRIBUTIONS

Medical image learning: Developed and evaluated a hybrid EfficientNet–Vision Transformer workflow for five-class diabetic-retinopathy classification.

Dataset engineering: Curated a 9,640-image retinal fundus research dataset from six sources: APTOS, DDR, DeepDRiD, EyePACS, IDRiD and MESSIDOR.

Explainability & validation: Applied Grad-CAM and SHAP to support transparent interpretation; the published diabetic-retinopathy study reported 97.23% test accuracy.

Cross-domain experimentation: Researched brain MRI localization/segmentation, otoscopic disease classification, plant disease analysis and NLP-assisted research workflows.

TECHNICAL AND ACADEMIC SKILLS

AI & deep learning

CNNs, ResNet, EfficientNet, transformers (ViT), hybrid models, ensemble learning, transfer learning.

Data science & evaluation

Python, preprocessing, augmentation, stratified splits, cross-validation, imbalance handling, calibration, classification metrics.

Explainable AI

Grad-CAM, SHAP, LIME, attention visualizations, model interpretation and error analysis.

Research engineering

TensorFlow/Keras, scikit-learn, OpenCV, pandas, NumPy; Jupyter, Kaggle and experiment tracking.

Scientific communication

Research design, literature synthesis, manuscript writing, reproducible reporting, LaTeX and MS Office.

Programming & web

C++, PHP, JavaScript, HTML, CSS, relational databases, web-based applications.

SELECTED PROJECTS

Clinic Management System | Developed a PHP, HTML, CSS and JavaScript portal for patient records, appointments, physician information and billing.

Hamza Shahbaz Artificial Intelligence Lab | Personal online research portfolio presenting project history and collaborative research interests.

LANGUAGES

Urdu (native) • **Punjabi** (mother tongue) • **English** (C1 listening/reading, B2 speaking, C2 writing)

Additional languages: German and Italian (A1–A2); Arabic (skill-dependent self-assessment).

PUBLICATIONS & RESEARCH OUTPUTS

13 journal articles | 8 published · 5 accepted / in production

1. **Diabetic retinopathy detection enhancement using deep learning algorithms on hybrid dataset** **[Published]**
   Shahbaz, H., & Ali, N. (2026). Journal of Umm Al-Qura University for Medical Science, 12, Article 17. • DOI: 10.1007/s44361-026-00022-8
   Contributions First author · Corresponding author · Literature review · Methodology · Experiments · Writing

2. **NeuroDEEP-CNN: A Multi-Task Attention Framework for Joint Classification, Localization, and Segmentation of Brain Tumors on MRI** **[Published]**
   Shahbaz, H. (2026). Advances in Artificial Intelligence Research, 6. • DOI: 10.54569/aair.1975659
   Contributions Sole author · Methodology · Model training · Validation · Writing & editing

3. **An Intelligent Research Paper Assistant: Integrated Retrieval, Explainable Domain Classification, and Research-Gap Ranking** **[Published]**
   Shahbaz, H. (2026). Advances in Artificial Intelligence Research, 6. • DOI: 10.54569/aair.1975838
   Contributions Sole author · NLP methodology · Experiments · Evaluation · Writing & review

4. **Ear Diseases Detection Using Machine Learning and Deep Learning Algorithms on Otoscopic Images** **[Published]**
   Hamza, M.M., & Sarnıklı, A.R. (2026). OneHealth Plus Journal, 4, 1–12. • Article
   Contributions First author · Corresponding author · Literature review · Experiments · Writing · Proofreading

5. **Five-Phase Writing (FPW): An Empirically Evaluated Framework for Ethical, Traceable, and Human-Governed Research Writing with Large Language Models** **[Published]**
   Shahbaz, H., Dad, B., & Saud, S. (2026). AI and Ethics, 6, Article 437. • DOI: 10.1007/s43681-026-01304-y
   Contributions First author · Corresponding author · Conceptualization · Methodology · Experiments · Final review

6. **ALO: Advanced Learning for Potato Disease Detection and Classification Using Deep Learning and Interpretability** **[Accepted]**
   Hamza Shahbaz. Turkish Journal of Agricultural Engineering Research. Article ready for issue.
   Contributions Methodology · Experiments · Explainability · Manuscript writing

7. **Predicting Rule-Based Dairy Milk Intake Recommendations Using Machine Learning: A Robustness, Calibration, and Explainability Study** **[Published]**
   Shahbaz, H., Khaliq, F. & Sarnikli, A. (2026). Journal of Umm Al-Qura University for Medical Science, 12, Article 46. • DOI: 10.1007/s44361-026-00058-w
   Contributions First author · Corresponding author · Methodology · Model development · Validation · Writing

8. **EfficientResNetFusion: Hybrid Deep Learning Architecture with Multi-Method Explainability for Guava Fruit Disease Classification** **[Published]**
   Shahbaz, H. (2026). Transactions on Computer Science and Applications, 3(1), 27–37.
   Contributions Sole author · Hybrid modeling · Experiments · Explainability · Proofreading

9. **A Qualitative Analysis of Coronary Heart Disease Using Machine Learning Algorithms** **[Published]**
   Shahbaz, H. (2026). OneHealth Plus Journal, 2026, 1–10.
   Contributions Sole author · Data analysis · Model evaluation · Validation

10. **PUCC: A Reproducible Cloud-Based Framework for Optimizing Deep Learning Input Pipelines and Multi-GPU Utilization** **[Accepted]**
    Hamza Shahbaz. Journal of Cloud Computing (Springer Nature). • DOI: 10.1186/s13677-026-01007-z
    Contributions First author · Methodology · GPU experiments · Validation · Writing · Proofreading

11. **Explainable Stacked Ensemble Learning for Lifestyle-Based Tea-versus-Coffee Preference Prediction** **[Accepted]**
    Hamza Shahbaz. Journal of Artificial Intelligence and Data Science (DergiPark); manuscript 2006686.
    Contributions Methodology · Ensemble modeling · Validation · Writing · Proofreading

12. **CNN Benchmarking for Ginger Health Status Classification Using Deep Learning** **[Accepted]**
    Hamza Shahbaz. Innovative Artificial Intelligence (DergiPark); manuscript 1982653.
    Contributions Methodology · CNN benchmarking · Experiments · Validation · Writing · Proofreading

13. **THYREC-LC: A Leakage-Controlled Machine Learning Benchmark for Differentiated Thyroid Cancer Recurrence Prediction** **[Accepted]**
    Hamza Shahbaz. Journal of Artificial Intelligence and Data Science (DergiPark); manuscript 2006743.
    Contributions Methodology · Leakage control · Experiments · Validation · Writing · Proofreading

ACADEMIC REFEREES

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SIGNATURE

